Q → LLM → Response

prompt

Conversation system

Q

A

Q

A

LLM → Doesn't have memory

Each request → append the past conversation in the prompt

Context out of limit

LLM → Context limit

memory → DB / RAM } Storage

RAM → memory → new Question + prompt

Memory → Session / Thread

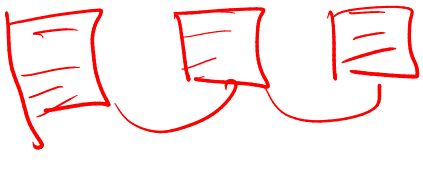
Short-term memory

→ memory within the same session



long-term memory

→ memory → Cross thread session



In-memory

⇒ Conversation are stored in temporary memory ⇒ RAM

→ Memory is cleared once session is terminated

persistant memory

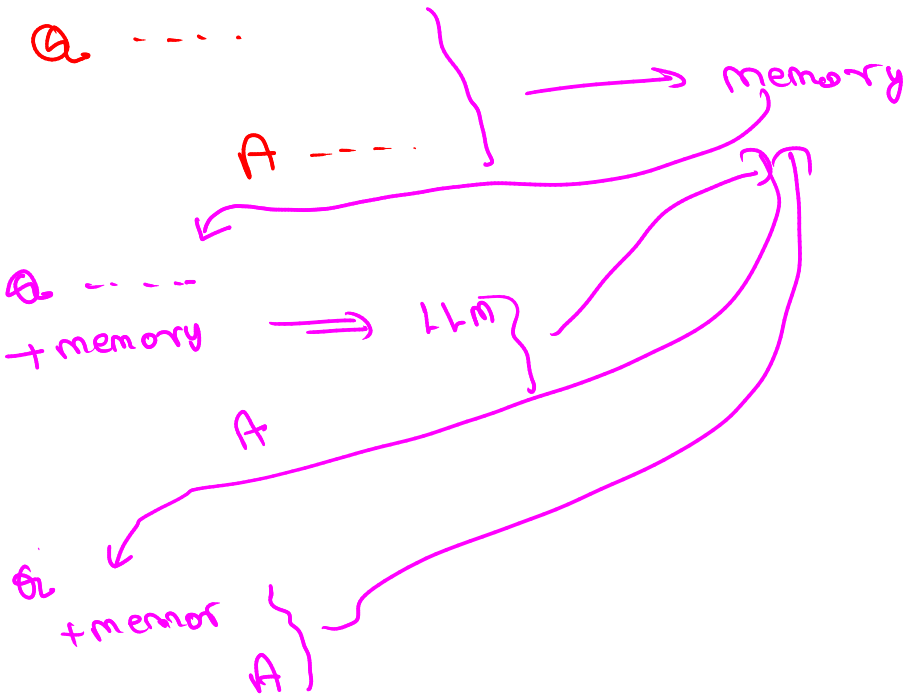
⇒ conversation are stored into in permanent memory

Database / File

→ memory is permanent

sessions → session id

↳ Inmemory → { }



persistant DB

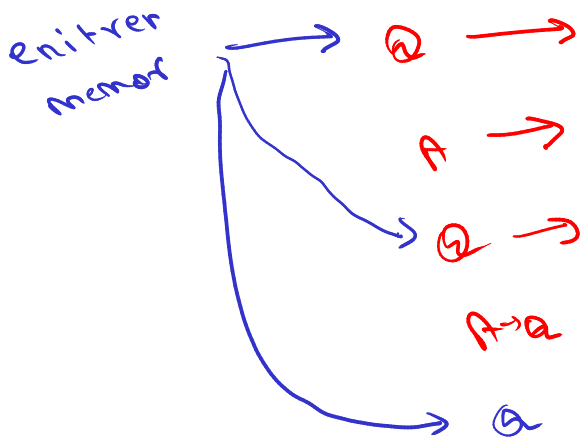
Local DB	cloud DB
.DB	Redis
SQL / sqlite	

Redis → key/value store DB

key : value

TTL → Time to Live

TTL - 5 min



LLM → context window limit

2. Token cost will increase

3. Context overload
   ↳ LLM performance will degrade

Optimation

⇒ Summarize conversation ✓ → How?
   ↳ LLM

⇒ logically remove unwanted message

open source model

GPT OSS 120B / 20B

llama scout

Qwen

GROQ API

closed soure model.

openAI API

Claude API

Gemini API

Download the model & use in the APP

CPU

Google colab

15GB GPU — 4-5 hrs

host a LLM model

Ollama

model → parameters — weights / Biases

Size

VRAM → usage is high

0.5B — SLM — small language models

1B

2B

1 Trillon

1B, 2B } 2 - 3 GB VRAM

7B, 8B } 8 - 9 GB GPU VRAM

13B, 20B } 15 - 25 GB GPU VRAM

25B

70B } → 80GB GPU VRAM

llama

Ollama → model hub ⇒ most famous open source models ⇓ optimized

Raw model

Quantization — pruning =

Quantized models

Quantization ⇒

train a LLM model

parameters weight

weight Biases ⇒ 0.8954

⇒ 0.8462

Type

Float32 ⇒ float 32 Bit

0.8462796 ⇒ 32 Bit

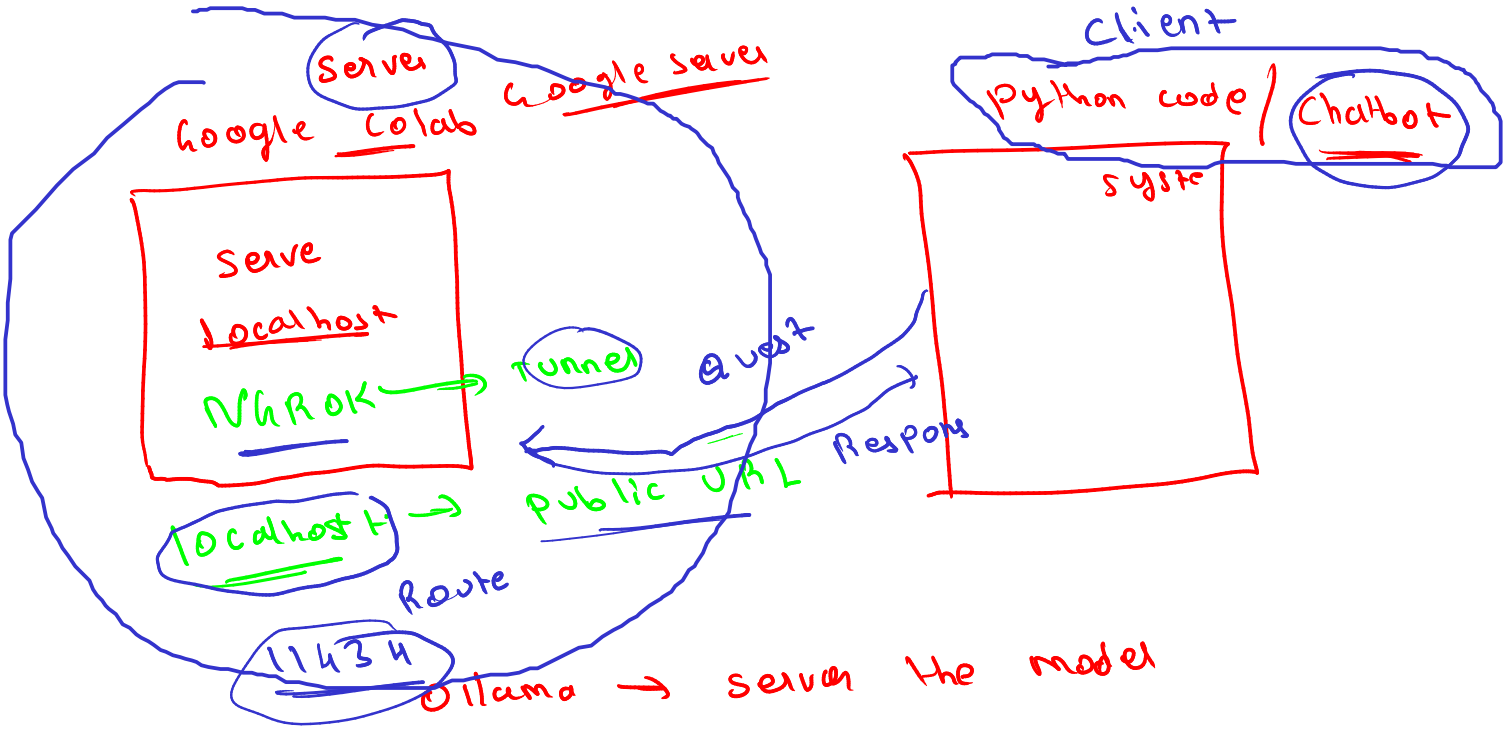
float16 ⇒ 16 Bit

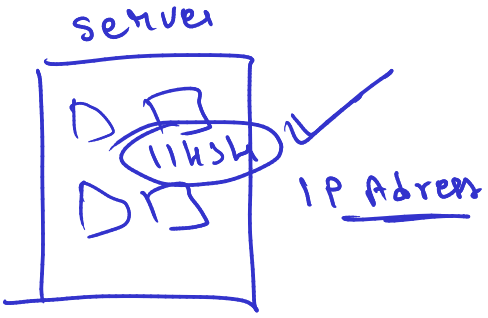
0.846 ⇒ 16 Bit

ollama → langchain

GPU → Google Colab – server

↳ 4-5 hrs





localhost → public URL

RAG